RW–10706

Sub. Code
731102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Education

ADVANCED EDUCATIONAL RESEARCH METHODOLOGY

(2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer ALL questions.

(5 × 15 = 75)

1. (a) Outline the need and characteristics of an educational research.

Or

(b) Distinguish the special features of types of research in education.

2. (a) Why do we need to incorporate the literature review in our research work?

Or

(b) Hypothesis-meaning, definition, types and characteristics – Discuss in detail.

3. (a) Method of study is the backbone of the research work – Support your answer.

Or

(b) Bring out the basic differences between survey and experimental research.

4. (a) Identify the characteristics of a good research tool.

Or

(b) Make out the basic differences between self-made and standardised tool.

5. (a) What are the uses of applying Wilcox on signed rank test and man whiney U test?

Or

(b) Multiple regression analysis and correlational analysis – List out the nature of their uses in inferring the data.

RW–10682

Sub. Code
421102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Tamil

ஆராய்ச்சி அணுகுமுறைகள்

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

கீழ்க்காணும் **அனைத்து** வினாக்களுக்கும் கட்டுரை வடிவில் விடையளிக்க.

1. (அ) அணுகு முறைகளைப் பின்பற்றிச் செய்யப்படும் ஆய்வுகள் சிறந்திருத்தலின் காரணங்களை ஆய்க.

(அல்லது)

(ஆ) விளக்கவியல்முறை பிற அணுகுமுறைகளுக்கு எவ்வாறு பங்களிப்பு செய்கின்றது?

2. (அ) யுங்கின் பங்களிப்புகள் இலக்கிய ஆய்வுகளுக்கு எவ்வாறு பயன்படும்?

(அல்லது)

(ஆ) சங்க இலக்கியப் பகுதிகளை உளவியல் ஆய்வு செய்து காட்டுக.

3. (அ) தமிழில் சமூகவியல் அணுகுமுறை பெற்றுள்ள வளர்ச்சியை மதிப்பீடு செய்க.

(அல்லது)

(ஆ) அமைப்பியல் அணுகுமுறையை விளக்குக.

4. (அ) அழகியலணுகுமுறையின் இயல்புகளை ஆய்க.

(அல்லது)

(ஆ) தமிழில் ஒப்பியலணுகுமுறை வளர்ச்சி பெற்ற முறையை ஆய்க.

5. (அ) வரலாற்றியல் அணுகுமுறையைத் தமிழிலக்கியத்தில் பொருத்தி விளக்குக.

(அல்லது)

(ஆ) மானிடவியல் அணுகுமுறை இலக்கியங்களைப் புரிந்து கொள்வதற்குப் பயன்படும் என்ற கூற்றை நிறுவுக.

RW–10683

Sub. Code
456101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Economics

RESEARCH AND STATISTICAL METHODS

(2013 Onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

All questions carry equal marks.

1. (a) Write a note on the data sources available for carrying out research in Economics.

Or

(b) Prepare a model questionnaire for making a study on "Impact of Alcohol consumption on village economy".

2. (a) What are the merits and demerits of the measures of dispersion?

Or

(b) Calculate give concentration ratio for the following data.

Number of Household	10	10	10	10	10	10	10	10	10	10
Values of assets owned in Rs. lakhs	2	10	60	100	300	800	1000	2000	8000	15000

3. (a) If $y = a + bx^2$, for the following data calculate a and b.

Y:	2	3	7	10	15	20	25	30
X:	3	10	15	18	20	22	28	40

Or

(b) Illustrate the methods of estimating trend and seasonal variations.

4. (a) Write a note of "Model Building".

Or

(b) Illustrate the uses of SPSS package.

5. (a) Discuss the stages of "Report writing".

Or

(b) Exemplify "citations".

RW–10684

Sub. Code
456102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015.

Economics

READING IN ECONOMIC THEORIES

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) Show the relation between cost and production curves.

Or

(b) What are the objectives of firms?

2. (a) Explain the keynesian method of output and employment determination.

Or

(b) Explain the working mechanism of "super multiplexer".

3. (a) Discuss the "Ranis-Fei model".

Or

(b) Illustrate "Cost Benefit Analysis".

4. (a) Discuss the theories of taxation.

Or

(b) Examine the problems of centre-state relations in India.

5. (a) IS International Trade an engine of growth? How?

Or

(b) What are the effects of changes in repo rate, CRR and SLR?

RW–10685

Sub. Code
921101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Library and Information Science

RESEARCH METHODOLOGY

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks. (5 × 15 = 75)

1. (a) Define Research. Explain its concept and characteristics of Research.

Or

(b) Mention the various types of Research and explain the significance of Applied Research.

2. (a) Explain Hypothesis. Discuss in detail about the types of Hypothesis.

Or

(b) What are the steps involved in formulation of a Research problem?

3. (a) Give a detailed account on Random sampling methods with examples.

Or

(b) What are the various methods of data collection? Explain questionnaire method of data collection.

4. (a) What are the various statistical tools and techniques used in library science research?

Or

(b) Enumerate the various steps involved in writing a Research Report.

5. (a) Give a detailed account on SPSS.

Or

(b) What is style manual? Discuss the MLA style manual with examples.

RW–10686

Sub. Code
921102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Library and Information Science

EMERGING THRUST AREAS IN LIBRARY AND INFORMATION SCIENCE

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) Define information communication technology. Explain its developments in libraries.

Or

(b) Give a detail account on librarianship 2.0.

2. (a) What is operating system? Discuss in detail about windows.

Or

(b) What are the various types of input devices.

3. (a) Mention the different types of house keeping operation in libraries.

Or

(b) Give a detail account on CDS/ISIS.

4. (a) Define network. Explain the various components of network and its use.

Or

(b) Explain in detail about network security measures.

5. (a) Define digital library. Discuss the need and characteristics of digital libraries.

Or

(b) Give an account on intellectual property right and its issues.

RW–10689

Sub. Code
519101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Zoology

RESEARCH METHODOLOGY

(CBCS 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) Write a detailed account on organizing and formatting a research project.

Or

(b) Explain the various methods of literature collection and mention its significance in biological research.

2. (a) Discuss the basic principle and applications of UV-Visible spectrophotometer.

Or

(b) Write an essay on good laboratory practices.

3. (a) Write the principle, components and applications of electrophoretic technique.

Or

(b) Give an account on the principle and applications of scanning electron microscopy.

4. (a) Explain the working principle and functions of GM counter.

Or

(b) Write an essay on southern hybridization and highlight its significance.

5. (a) Discuss the importance of correlation and regression analysis in biological research by citing suitable example.

Or

(b) Explain the types and significance of ANOVA in the field of biological research.

RW–10690

Sub. Code
519102

M.Phil. (Zoology) DEGREE EXAMINATION, DECEMBER 2015

ADVANCED ZOOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) Write a detailed account on Mendelian laws of inheritance with examples.

Or

(b) Explain the different aspects of cell cycle. Add a note on control and check points in cell cycle.

2. (a) Write an essay on Immunoglobulins.

Or

(b) Explain the biological significance of proteins.

3. (a) Write a detailed account on pathology and lab diagnosis of *Wuchereria bancrofti*.

Or

(b) Describe the life cycle and preventive measures of malarial parasite.

4. (a) Write an essay on integrated pest management.

Or

(b) Describe the life cycle, rearing methods and uses of silkworm and honey bees.

5. (a) Explain the role of micro and macroorganisms on bioremediation of Polyaromatic Hydrocarbons (PAH).

Or

(b) Write about the role of microorganisms in the degradation of oil spills.

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RW–10691

Sub. Code
561101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Computer Science

RESEARCH METHODOLOGY

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** the questions.

All questions carry equal marks.

$(5 \times 15 = 75)$

1. (a) (i) Describe the mathematical tools available for research.

(ii) Explain the importance of primary data.

Or

(b) (i) Describe the different types of research.

(ii) With suitable example, explain how can you measure skewness.

2. (a) (i) What are the criteria available for judging algorithms? Explain.

(ii) Describe the applications of stack.

Or

(b) (i) Write the steps in development of an algorithm and explain.

(ii) Compare the singly linked list with doubly linked list with examples.

3. (a) What do you mean by backtracking? Explain the eight questions problem and write an algorithm for the same. Also analyse your procedure.

Or

(b) Explain the Dijkshtra's algorithm to find the shortest path. Develop an algorithm and analyse your procedure.

4. (a) (i) What is the output plain text

hello, my name is Sachin Ramesh Tendulkar

if we use Caesar Cipher to encode it?

(ii) Describe the steps in DES.

Or

(b) (i) Explain the RSA algorithm with an example.

(ii) What are the limitations of a firewall? Explain.

5. (a) Describe the different tools available for writing and publishing the thesis.

Or

(b) Discuss about

(i) Page and chapter format

(ii) Footnotes.

RW–10694

Sub. Code
581102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Physics

ADVANCED PHYSICS

(CBCS 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

All questions carry equal marks.

1. (a) (i) Describe Fock space and harmonic oscillator based on non-relativistic theory. (8)

(ii) Explain in detail the quantum field operator and its applications. (7)

Or

(b) (i) Obtain the Klein Gordon equation for scalar field. (8)

(ii) Derive the fundamental equation of motion for scalar meson field. (7)

2. (a) (i) Write a note on non-localized bonds. (6)

(ii) What is hybridization? Explain Sp^2 and Sp^3 hybridization with example. (9)

Or

(b) (i) Discuss few of the self-consistent field techniques. (6)

(ii) Explain the elementary ideas of Hartree and Fock Methods? (9)

3. (a) (i) Briefly discuss the solid state laser with examples. (8)

(ii) Give the merits and applications of gas lasers. (7)

Or

(b) (i) Explain the working principle of liquid dye laser with suitable example. (8)

(ii) Write the merits and demerits of chemical laser over the solid state lasers. (7)

4. (a) (i) Discuss the basic principles and instrumentation of Raman spectrometer. (8)

(ii) Distinguish merits and demerits of FT-IR and Raman spectroscopy. (7)

Or

(b) (i) Briefly discuss the working principle and applications of electron spin spectrometer. (8)

(ii) Give the advantages of NMR over ESR. (7)

5. (a) (i) Explain the principles and working of XPS. (8)

(ii) What are the electrochemical parameters usually analysed using XPS. (7)

Or

(b) (i) State principle and applications of Differential Thermal Analyzer. (6)

(ii) Write a note on DSC with suitable diagram. (9)

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RW–10695

Sub. Code
596101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Chemistry

RESEARCH METHODOLOGY IN CHEMISTRY

(CBCS 2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Explain the specific roles of journals, periodicals, monographs, books, treatises, encyclopedia, abstracts and patents in providing scientific informations.

Or

(b) Describe how a research problem should be planned and indicate how a complete historical survey regarding the research problem can be achieved.

2. (a) Give a detailed account of all the chromatographic techniques in successful identification, separation and purification of organic compounds.

Or

(b) (i) Discuss the principle of colourimetry.

(ii) Explain the instrumentation features of flame photometry.

(iii) Describe the applications of atomic absorption spectroscopy.

3. (a) (i) Explain the important applications of EPR spectroscopy.

(ii) An organic compound with molecular formula $C_5H_{10}O$ has five ^{13}C NMR signals at 142 ppm, 112 ppm, 60 ppm, 41 ppm and 22 ppm. The ^{1}H NMR spectrum (Chemical sifts in δ scale) has signals at 4.8 ppm (1 H, broad), 4.7 ppm (1 H, broad), 3.7 ppm (2H, triplet), 2.2 ppm (2H, triplet), 2.1 ppm (1 H, broad) and 1.8 ppm (3H, singlet). Assign the structure.

Or

(b) (i) Write a note on Auger electron spectroscopy.

(ii) An organic compound with molecular formula $C_9H_9BrO_2$ has an IR band at 1725 cm^{-1}. The 1H NMR spectral data in δ scale is given: 8.2, triplet, 1H; 8.0, doublet of triplet, 1H; 7.6, doublet of doublet of doublet, 1H; 7.2, triplet, 1H, 4.5, quartet, 2H, 1.5, triplet, 3H. Assign the structure of the compound.

4. (a) Discuss the broad applications of XRD, TG/DTA and AFM.

Or

(b) Write notes on SEM and TEM.

5. (a) (i) Explain how errors are classified.

(ii) Distinguish between accuracy and precision.

(iii) Establish the importance of significant numbers with adequate examples.

Or

(b) Write notes on:

(i) Students 't' test

(ii) Mean and standard deviation

(iii) Correlation coefficient.

RW–10696

Sub. Code
596102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Chemistry

AREA OF SPECIALIZATION IN CHEMISTRY

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** the questions. (5 × 15 = 75)

1. (a) Explain the electro chemical reactions, characteristics and uses of the following batteries

   (i) Leclanche cell.

   (ii) Alkaline batteries.

   (iii) Water activated batteries.

   (iv) Ni-Metal hydride batteries.

   (v) Lithium-ion batteries.

Or

(b) (i) How do fuel cells differ from batteries?

(ii) Explain the following fuel cell designs.

(1) Alkaline fuel cell system.

(2) Molten carbonate fuel cell system.

(3) Solid oxide fuel cell system.

(4) PEM oxide fuel cell system.

2. (a) Discuss in detail about the mechanism and kinetics of free radical, cationic and anionic polymerisations.

Or

(b) Write note on the following :

(i) Determination of glass transition temperature.

(ii) Factors affecting Tg and crystallinity of solids.

(iii) Measurement of Tg by DSC, DTA and TMA.

3. (a) How are nanomaterials classified? Give examples. Describe the different startergies for the synthesis of (Zero) OD and ID Nanomaterials.

Or

(b) How are 2D and 3D nanomaterials synthesized by different startergies? Discuss the technical applications.

4. (a) How are the following techniques used for the characterization of nanomaterials? Explain with examples.

(i) Small angle x-ray scattering.

(ii) Scanning Electron miscorscopy.

(iii) Low energy electron difraction.

(iv) Atomic force microscopy.

(v) ESCA.

Or

(b) Explain the following characterization techniques of nanomaterials with example.

(i) Transmission Electron microscopy.

(ii) UV photo electron spectroscopy.

(iii) Infrared spectroscopy.

(iv) XPS

(v) Thermal Analysis method.

5. (a) Discuss the applications of nanomaterials in the following fields with suitable examples.

   (i) Sensors

   (ii) Energy devices

   (iii) Fuel cells and Electrical devices.

Or

   (b) How are nanomaterials used in Dye sensitized solar cells, pharmaceuticals, pollution control, food and packaging? Explain in detail with example.

RW–10697

Sub. Code
571101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Mathematics

RESEARCH METHODOLOGY FOR ALGEBRA

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

All questions carry equal marks.

(5 × 15 = 75)

1. (a) (i) Define cyclic group. Prove that direct product of two cyclic group is cyclic.

(ii) Prove that there are two non-abelian groups of order 8.

Or

(b) (i) Define abelian tower. Construct an abelian tower for $G = GL(n,k)$.

(ii) State and prove Butterfly lemma.

2. (a) (i) Let B be a subgroup of a free abelian group A. Prove that B is also free and any two bases of B have the same cardinality.

(ii) Let A be a finitely generated torsion-free abelian group. Prove that A is free.

Or

(b) (i) Define category and write a detailed on various types of categories and functor.

(ii) Show that coproducts exist in the category of groups.

3. (a) (i) Identify all maximal ideals in the ring of integers Z.

(ii) Describe the construction of group ring $A(G)$ from a group G.

Or

(b) (i) Let P be a prime ideal of a commutative ring A. Prove that $A_P = S^{-1}A$ is a local ring.

(ii) Let I and J be two ideals of A. Prove that $S^{-1}(I+J) = S^{-1}I + S^{-1}J$, $S^{-1}(I \cap J) = S^{-1}I \cap S^{-1}J$ and $S^{-1}(IJ) = (S^{-1}I)(S^{-1}J)$.

4. (a) (i) Prove that a sequence

$X' \xrightarrow{\lambda} X \longrightarrow X'' \longrightarrow 0$ is exact if and only if

$$Hom_A(X',Y) \to Hom_A(X,Y) \to Hom_A(X'',Y) \to 0$$

is exact.

(ii) Show that $Hom_A(N, M \times M') \cong Hom_A(N,M) X Hom_A(N,M')$ as abelian groups.

Or

(b) (i) Let $0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \longrightarrow 0$ be an exact sequence of modules. Prove that, if there exists a $\phi : M'' \to M$ such that $g \circ \phi = id$, then

$M = \ker g \oplus \operatorname{Im} \phi$, $M \cong M' \oplus M''$.

(ii) Let A be a ring and P a module. List out the four equivalent conditions for P to be a projective module. Prove the equivalence of the four conditions.

5. (a) Let E be a finitely generated torsion module prove that $E \cong \frac{R}{(q_1)} \oplus ..... \oplus \frac{R}{(q_r)}$ where $q_1, ..., q_r$ are non-zero elements of R and $q_1 | q_2 | ..... | q_r$. Further prove that the sequence of ideals $(q_1), ..., (q_r)$ is uniquely determined by the above condition.

Or

(b) State and prove Hilbert's theorem.

RW–10698

Sub. Code
571102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Mathematics

FUNCTIONAL ANALYSIS

(2015 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** the questions.

All questions carry equal marks.

$(5 \times 15 = 75)$

1. (a) (i) Prove that every locally compact topological vector space X has a finite dimension.

(ii) If Y is a subspace of a topological vector space X, and Y is an F-space. Then prove that Y is a closed subspace of X.

Or

(b) (i) Prove that a set E in a topological vector space is bounded if and only if every countable subset of E is bounded.

(ii) Prove that the spaces $C(\Omega)$ is not locally bounded.

2. (a) State and prove open mapping theorem.

Or

(b) If X, Y, Z are Banach spaces and $B : X \times Y \to Z$ is bilinear and continuous, prove that there exists $M < \infty$ such that $||B(x,y)|| \le M||x||\,||y||$, $(x \in X, y \in Y)$. Is completeness needed here?

3. (a) If V is a neighbourhood of 0 in a topological vector space X and if $K = \{\wedge \in X^* : |\wedge x| \leq 1 \; for\, every\, x \in V\}$ then prove that K is weak* compact.

Or

(b) (i) If K is a compact set in a locally convex space X, and if $\overline{co}(K)$ is also compact, then prove that every extreme point of $\overline{co}(K)$ lies in K.

(ii) Suppose that P is an extreme point of some convex set K, and that $P = t_1x_1 + ... + t_nx_n$, where $\sum t_i = 1, t_i > 0$ and $x_i \in K$ for all i, then prove that $x_i = p$ for all i.

4. (a) (i) Prove that $T \in \mathfrak{W}(X, Y)$ is an isometry of X *onto* Y if and only if T^* is an isometry of Y^* *onto* X^*.

(ii) Let σ and τ be the weak* topologies of X^* and Y^* respectively, and prove that S is continuous linear mapping of (Y^*, τ) into (X^*, σ) if and only if $S = T^*$ for some $T \in \mathfrak{W}(X, Y)$.

Or

(b) (i) If $T \in \mathfrak{W}(X, Y)$ and $R(T^*) = N(T)^{\perp}$, prove that $R(T)$ is closed.

(ii) Assumer $T \in \mathfrak{W}(X, Y)$ and $T(X) = Y$, show that there exists $\delta > 0$ such that $S(X) = Y$ for all $S \in \mathfrak{W}(X, Y)$ with $||S - T|| < \delta$.

5. (a) (i) Prove that $D(\Omega)$ has the Heine-Borel properties.

(ii) If E is an arbitrary closed subset of R^n, show that there is an $f \in C^\infty(R^n)$ such that $f(x) = 0$ for every $x \in E$ and $f(x) > 0$ for every other $x \in R^n$.

Or

(b) Find all distribution $u \in D'(R^n)$ that satisfies at least one of the following two conditions.

(i) $\tau_x u = u$ for every $x \in R^n$

(ii) $D^\alpha u = 0$ for every α with $|\alpha| = 1$.

RW–10699

Sub. Code
642101/661101/ 671101/662101/ 681101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Management

BUSINESS RESEARCH METHODS/METHODOLOGY OF BUSINESS RESEARCH

(Common for Mgt./C.S./B&I/I.B./Commerce)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (7 × 3 = 21)

Answer **all** questions.

All questions carry equal marks.

1. Distinguish between action research and experimental research.

2. What is 'test of validity'?

3. What is sampling distribution?

4. Distinguish between questionnaire and schedule.

5. What do you mean by 'editing of data'?

6. Write a short note on 'footnote'.

7. What is Q-sort?

Section B (4 × 6 = 24)

Answer any **four** questions.

All questions carry equal marks.

8. What is case study research?

9. What are the possible sources of error in measurement?

10. What are the determinants of sample size?

11. What are the principles that govern the sequence of questions on a schedule?

12. Briefly explain the documentary sources of data.

13. Examine the different documentation styles used in Research.

Section C (2 × 10 = 20)

Answer **all** questions.

All questions carry equal marks.

14. (a) What kind of literature should a researcher review and for what purpose?

Or

(b) Explain the steps involved in construction of Likert Scale.

15. (a) What are merits and demerits of personal interview compared to questionnaire?

Or

(b) Explain the logical and statistical methods of analyzing data.

Section D **(1 × 10 = 10)**

(Compulsory)

16. Read the following case and answer the questions.

Dharshan Dog Foods, a new dog food, made a good start in its first year on the market, achieving a market share of about 8 percent. Ambitious plans were underway for the second year plans that include a strong TV advertising program. The advertising agency had come up with a new campaign theme.

Dharshan Dog Food aimed at the quality and of the market and the price reflected this fact, although the management believed the price was no more than average when quality was considered.

A low cost draft of a proposal TV commercial was developed on videotape. Families in each of the 3 major cities who owned a dog were recruited by telephone. The individual in the family who selected the food fed to the dog was invited to visit a research office in a regional shopping centre and a specific appointment was made. Such appointments were made with 40 individuals in each city, with the expectation that this would produce 35 interviews per city.

The consumers were told they were going to see some rough commercials after which they would be asked questions. All 5 commercials were shown, and the respondents were then asked what products and what brands had been shown in the commercials.

Critically analyze the attitude measurement methodology in this case.

RW–10700

Sub. Code
642102/671102/ 661102/662102/ 681102

M.Phil. (Management) DEGREE EXAMINATION, DECEMBER 2015

STATISTICAL TECHNIQUES OF RESEARCH/ TECHNIQUES OF BUSINESS RESEARCH

(Common for Mgt./CS/B&I/IB/Commerce)

(2012 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A ($7 \times 3 = 21$)

Answer **all** questions.

All questions carry equal marks.

1. How would you overcome the disadvantages of measures of central tendency?

2. Describe the significance of regression.

3. Distinguish between type I and type II errors.

4. What useful purpose does pot hoc tests in ANOVA serve?

5. What is 'Run Test'?

6. What is heteroscedasticity?

7. Enumerate the uses of chi-square test.

Section B (4 × 6 = 24)

Answer any **four** questions.

All questions carry equal marks.

8. Differentiate between inductive and deductive methods.

9. The following table gives the age of cars of certain make and annual maintenance costs.

X:	40	50	38	60	65	50	35
Y:	38	60	55	70	60	48	30

Obtain regression equation for y on x.

10. Explain the significance of Yates correction in estimating inter-relationship among variables.

11. Three demand values of a product over the past 10 years are summarized in the given table. Find the auto-correlation coefficients with one year lag, two year lag and three year lag.

Year	Demands (in '000')	Year	Demands (in '000')
1	20	6	90
2	30	7	100
3	50	8	120
4	60	9	140
5	80	10	160

12. What is factor rotation?

13. What do you mean by correlation analysis? State the properties of coefficient of correlation.

Section C

$(2 \times 10 = 20)$

Answer **all** questions.

All questions carry equal marks.

14. (a) List and explain different data presentation techniques.

Or

(b) There are three main brands of a certain powder. A set of 120 sample values is examined and found to be allocated among four groups (A, B, C and D) and three brands (I, II, and III) as shown here under

Brands	Groups			
	A	B	C	D
I	0	3	8	15
II	5	8	13	6
III	8	19	11	13

Is there any significant difference in brands preference? Answer at 5% level, using one-way ANOVA.

15. (a) List and explain different clustering techniques.

Or

(b) Enumerate the applications, advantages and disadvantages of conjoint analysis.

Section D $(1 \times 10 = 10)$

(Compulsory)

16. In an experiment to determine which of three different missile systems is preferable, the propellant burning rates were measured. The data after coding is given below.

System I	24.0	16.7	22.8	19.8	18.9			
System II	23.2	19.8	18.1	17.6	20.2	17.8		
System III	18.4	19.1	17.3	17.3	19.7	18.9	18.8	19.3

Use the Kruskal-Wallis test and a significance level of $\alpha = 0.05$ to test the hypothesis that propellant burning rates are the same for the three missiles.

RW–10701

Sub. Code
761101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

English

RHETORIC AND RESEARCH METHODOLOGY

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer ALL questions.

($5 \times 15 = 75$)

1. (a) Analyze the five parts of discourse.

Or

(b) Discuss the different kinds of fallacies.

2. (a) Bring out the various methods of exposition.

Or

(b) Examine the techniques of argument, description, and narration.

3. (a) How would you select a research topic?

Or

(b) Examine the importance of note taking.

4. (a) Discuss the language and style used in research writing.

Or

(b) Distinguish between MLA style and APA style.

5. (a) Analyze the mechanics of writing.

Or

(b) Bring out the layout of a thesis.

RW–10702

Sub. Code
761102

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

English

LITERARY CRITICISM : CONTEMPORARY CRITICAL THEORIES

(2013 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Define the literary terms: Ambiguity, Phenomenology, Irony, Semiotics and Humanism.

Or

(b) Explain the terms: Affective Fallacy, Denotation, Expressionism, Hermeneutics and Stylistics.

2. (a) Explore the tenets of formalistic approach.

Or

(b) What are the chief features of psychological approach?

3. (a) How does Gennette argue that structuralism is more than a literary exercise?

Or

(b) How far is "Structure, Sign and Play in the Discourse of Human Sciences" a crusade against the traditional and metaphysical interpretation of texts?

4. (a) How does Barthes establish that a text is the tissue of quotations?

Or

(b) Examine the emphasis on the role of "interpretive community" as elaborated in Fish's "Is There a Text in This Class".

5. (a) Examine the three phases that Showalter presents in "Towards a Feminist Poetics".

Or

(b) Consider *A Room of One's Own* a sustained polemic on the power of privilege, confidence and entitlement.

RW–10705

Sub. Code
731101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Education

INSTRUCTIONAL TECHNIQUES

(2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer ALL questions.

(5 × 15 = 75)

1. (a) Tabulate the educational implications of Humanism, secularism and democracy in detail.

Or

(b) Education is for achieving social order in the society. Justify.

2. (a) Describe Maslow's theory of self-actualization in detail.

Or

(b) Discuss the educational contribution of piaget and Bruner.

3. (a) Explain the components and relevance of system approach to education.

Or

(b) How would you effectively use programmed instruction in your classroom?

4. (a) Describe any one model from social interaction family.

Or

(b) Describe any one model from a behaviour intraction models of teaching.

5. (a) Teacher is a problem solve justify elaborately.

Or

(b) Explain the important professional enrichment programmes for the would be teachers.

RW–10681

Sub. Code
421101

M.Phil. DEGREE EXAMINATION, DECEMBER 2015

Tamil

ஆராய்ச்சி நெறிமுறைகள்

(CBCS – 2013 onwards)

Time : 3 Hours Maximum : 75 Marks

கீழ்க்காணும் அனைத்து வினாக்களுக்கும் கட்டுரை வடிவில் விடையளிக்க.

(5 × 15 = 75)

1. (அ) ஆய்வுச் சிக்கலைத் தெரிவு செய்யும் முறையை விவரிக்க.

(அல்லது)

(ஆ) ஆய்வாளனின் தகுதிகளையும் பண்புகளையும் விளக்குக.

2. (அ) ஆய்வு மூலங்களைத் தொகுக்கும் முறையை விளக்குக.

(அல்லது)

(ஆ) கருதுகோளின் வகைகளையும் உருவாக்கும் முறையையும் விளக்குக.

3. (அ) களப்பணி உத்திமுறைகளை விளக்குக.

(அல்லது)

(ஆ) நூலகம் வழித் தரவுகள் திரட்டும் முறையை விளக்குக.

4. (அ) ஆய்வேட்டின் அமைப்பை விளக்குக.

(அல்லது)

(ஆ) ஆய்வேட்டை உருவாக்கும் போது கையாள வேண்டிய நெறிமுறைகளை விளக்குக.

5. (அ) வரலாற்றியல் முறையைப் பயன்படுத்தும் முறையை விளக்குக.

(அல்லது)

(ஆ) ஆய்வேட்டில் சான்றாதாரங்கள் பயன்படுத்தும் முறையை விளக்குக.